

VHDL-Based Mini NPU for Logistic Regression Inference on FPGA

- **About the system**

The system takes four binary input features from FPGA switches, multiplies them by predefined signed weights, adds a bias value, and computes a weighted sum. This sum is then passed to a simple threshold activation unit, which classifies the input as class 0 or class 1. The predicted class is displayed using an LED, while the calculated score can be shown on a seven-segment display.

- **Architecture diagram**

